WATER RESOURCES, WELLS & PUMPS · NOTE PU02

Pump Principles and Reciprocating Pumps — Hand, Treadle, Deep-Well and India Mark II

Course topics: FMPE8.6 L6, CAE1.3 L15

Why pumps

Motor- or engine-driven pumps dominate large lift-irrigation schemes because high outputs are easy to reach and control, and mechanical efficiency and economy matter in large installations.

Four principles of pumping

1. Atmospheric pressure
2. Centrifugal force
3. Positive displacement
4. Movement of fluid columns caused by differences in specific gravity

Classification by mechanical principle

Class	Types
Displacement pumps	Reciprocating — lift (suction) pump; force pump, single- or double-acting. Rotary pump
Centrifugal pump	—
Turbine pumps	Deep-well turbine; submersible
Propeller pumps	Axial flow; mixed flow
Airlift pump	—

Pumping conditions decide the choice. Centrifugal and submersible turbine pumps are the most common in irrigation.

Reciprocating pumps

- **Parts:** cylinder, airtight plunger or piston, piston (pump) rod, cup-shaped leather packing, inlet valve at the bottom of the cylinder and discharge valve (usually in the plunger).
- **Working:** pushing the plunger in forces water (or air) out; withdrawing it creates a vacuum that draws water in. Neglecting leakage, discharge equals the volume displaced — hence **positive (displacement) pumps**.
- **Performance:** high heads but small capacity, so mostly used for domestic supply.

Shallow-well (lift-type) pumps

- They raise water by atmospheric pressure. The common hand pump is the **pitcher pump**: head and cylinder cast in one piece, standing on the ground with a suction pipe into the well.
- At an atmospheric pressure of 1 kg/cm² the theoretical lift is about 10 m; friction and imperfections keep the practical lift within 8 m.

Treadle pump

- A human-powered (leg-driven) suction pump, popular where the water level is about 10 m or less; RuTAG, IIT Delhi, has developed improved versions — one built from local plumbing and hand-pump parts, and an improved existing treadle pump.
- **Working:** raising the piston creates a vacuum that sucks water up the pipe through a non-return valve (which stops backflow); pushing it down forces water out through a small valve; repeated strokes fill a channel or tank.
- **Improved design:** two cylinders side by side with pistons linked by a rope over a pulley, so one goes down as the other comes up; the rhythm makes it easy to accept. A valve closing on the down-stroke keeps pressure in the delivery pipe, so it can run sprinklers or drippers or fill a header tank.

Pump	Delivery
Single-acting	One discharge stroke in every two strokes; intermittent; suction and delivery at one end of the cylinder
Double-acting	Delivers on every stroke; suction and delivery at both ends; usually a shallow-well pump, practical where the water level is < 7 m

Deep-well plunger pumps

- The cylinder with plunger and valves sits at the bottom of the well, so water can be lifted to almost any height; the plunger is worked by a handle or crankshaft.
- Hand operation suits depths up to 45 m; deeper, the weight of the water column and connecting rod makes pumping hard. They are usually single-acting, slow and need high starting power.

India Mark II deep-well hand pump

- Developed with Richardson & Cruddas (1972) Ltd., a Government of India undertaking; widely used since the 1980s.
- Same basic features as a deep-well plunger pump; the cylinder assembly has 2 or 3 pump buckets depending on depth; for extra-deep water a handle of 40 cm diameter and counterweights are provided.
- Choose it where the dry-season groundwater level is below 6 m.
- **Installation:** close the well top with a removable reinforced-concrete slab to keep pollutants out; for bore or shallow tube wells build a permanent concrete or masonry platform of about 1 m^2 with drainage.
- **Mechanically operated reciprocating pumps** convert rotary motion through a crankshaft (working head). Their many moving parts and wear make them unsatisfactory for irrigation; they are used with windmills and as mud pumps on rotary drilling rigs.

Trouble (deep-well plunger pump)	Cause	Remedy
Water runs back down the piping	Defective valves or an obstruction below	Remove the valves; clean the seats
Air bubbles in the discharge	Leak or split in the suction pipe	Stop the leak; replace the pipe
Pump runs light, no discharge	Rod joint disconnected	Open the pipe system; tighten the joints
Obstruction on the upstroke	Rod joints loose, rod lengthened	Open the system; tighten the joints

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). Lessons on pumps (classification, centrifugal and turbine pumps, pumping power). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Reciprocating pumps are called positive pumps because

(a) they run on positive pressure
(b) discharge equals volume displaced
(c) they never leak
(d) they need priming

Q2. Practical suction lift of a shallow-well hand pump does not exceed about

(a) 4 m
(b) 8 m
(c) 15 m
(d) 30 m

Q3. A double-acting pump delivers water

(a) every other stroke
(b) every stroke
(c) only on suction
(d) only at start

Q4. Hand-operated deep-well plunger pumps suit depths up to about

(a) 7 m
(b) 15 m
(c) 45 m
(d) 150 m

Q5. India Mark II hand pumps are chosen where the dry-season water level is below

(a) 1 m
(b) 6 m
(c) 30 m
(d) 60 m

Answers

Q1 — (b) discharge equals volume displaced
Q2 — (b) 8 m — *Theoretical ≈ 10 m at 1 kg/cm².*
Q3 — (b) every stroke
Q4 — (c) 45 m
Q5 — (b) 6 m